

BRANDON HAN

Software Engineer | Specializing in AI/ML, Automated Systems, and Backend Development

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TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C, JavaScript, Typescript, , SQL, Go  
Backend / Databases: FastAPI, Flask, Node.js, PostgreSQL, MySQL, MongoDB, REST APIs  
AI / Data / ML: Computer Vision, OpenAI APIs, PyTorch, TensorFlow, Hugging Face, scikit-learn, Pandas, NumPy  
Cloud & DevOps: AWS (S3, ECS, Fargate, ECR, EC2, Bedrock, Textract), Docker, Git, CI/CD, React

EXPERIENCE

**Wasabi Cloud Technologies** Boston, MA  
**AI Engineering Intern** June 2025-August 2025

- Engineered an automated support-triage tool via **FastAPI**, leveraging **RAG architecture** and **OpenAI APIs** to retrieve relevant case information and generate structured diagnostic reports, reducing weekly triage meeting times by 33%
- Processed support tickets and internal documents using **Pandas** to construct labeled datasets for downstream LLM fine-tuning

**California NanoSystems Institute** Los Angeles, CA  
**Research Operations Manager & Lead Automation Engineer** January 2019-December 2023

- Integrated robotic handling using **computer vision** and **low-latency C++ hardware interfaces** to support automated experiment workflows for over 200+ research customers via modular workflows and pipelines that involve automated liquid handling, microscopes, and temperature-sensitive storage
- Deployed **PyTorch** and **TensorFlow** inference on GPU-backed **Docker** containers that analyzed approximately 1,500 images at full resolution per run in 20 minutes
- Developed and customized **Python**-backed software workflows using **segmentation, object detection, and classical CV** to support customer-specific cell-screening and pharmacology workflows
- Secured \$1.2M in infrastructure funding through grant proposals and contributed to 5 peer-reviewed publications involving ML, imaging, and pharmacology computations

PROJECTS

**Bias Induced News Generation with LLMs** Los Angeles, CA  
**Software Engineer** May 2026

- Built a reproducible evaluation pipeline deployed on **AWS EC2** with **PyTorch**, and **Hugging Face** to measure how politically skewed fine-tuning data affected news generated by multiple LLMs (**Mistral, GPT-4o, Deepseek, Qwen**)
- Developed data preprocessing and finetuning pipelines, and trained a custom RoBERTa classifier to evaluate generated texts and score analytical errors

**Accessibility-Focused AI Chrome Extension** Los Angeles, CA  
**Project Manager / Engineer** February 2026

- Built and deployed a **React** and **JavaScript** Chrome Extension in 24 hours that extracted webpage content, generated summaries, and answered page-grounded questions through **Node.js REST APIs** connected to GPT-4o model.
- Implemented read-aloud functionality with the **Web Speech API** and keyboard shortcuts/hot-keys through the **Chrome Commands API** to support an accessible browsing experience

**Stock Sentiment Tracker** Los Angeles, CA  
**Software Engineer** November 2025

- Fine-tuned FinBERT via **Hugging Face**, adapting it for informal finance text using Twitter-RoBERTa and training on social-market language filled with emojis and other modern terms
- Built **Flask** endpoints to serve sentiment predictions and used **SHAP** to identify which words influenced model outputs

EDUCATION

**University of Southern California** Los Angeles, CA  
**Master of Science, Computer Science** January 2024-May 2026  
Relevant Courses: Advanced Databases | Algorithms | Machine Learning | Deep Learning & Optimization | Web Technologies

**University of California, Los Angeles** Los Angeles, CA  
**Bachelor of Science, Biochemistry** September 2017-December 2019